

Lecture 13: Constitutive theory

David Al-Attar

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Outline and Motivation

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- Use the principle of material frame indifference to find the precise form of the strain energy function, W .
- Derive the **linearised equations of motion** for small deformations around a stress-free state. These are crucial for seismology.
- Discuss **material symmetries** (e.g., isotropy) to simplify the equations for real materials.

Material Frame Indifference Revisited

Recall that the strain energy, W , must be independent of superimposed rigid body rotations.

Mathematical Condition

For any rotation matrix $\mathbf{Q}$, the strain energy function must satisfy:

$$W(\mathbf{x}, \mathbf{Q}\mathbf{F}) = W(\mathbf{x}, \mathbf{F})$$

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To understand what this means for the form of W , we need a key result from linear algebra.

The Polar Decomposition Theorem

The Theorem

Any invertible matrix $\mathbf{F}$ (with $\det \mathbf{F} > 0$) can be uniquely decomposed as:

$$\mathbf{F} = \mathbf{R}\mathbf{U}$$

where:

- $\mathbf{R}$ is a rotation matrix (the rotation part).
- $\mathbf{U}$ is a symmetric, positive-definite matrix (the stretch part).

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Geometric Interpretation

Any deformation can be seen as a pure stretch ($\mathbf{U}$) followed by a rigid rotation ($\mathbf{R}$).

Note: The proof of this theorem is non-examinable.

Implications for the Strain Energy (1/2)

Let's apply the polar decomposition, $\mathbf{F} = \mathbf{R}\mathbf{U}$, to the frame indifference condition:

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The principle states that for any rotation $\mathbf{Q}$, $W(\mathbf{x}, \mathbf{Q}\mathbf{F}) = W(\mathbf{x}, \mathbf{F})$. If we choose $\mathbf{Q} = \mathbf{R}^{-1}$, we find:

$$W(\mathbf{x}, \mathbf{F}) = W(\mathbf{x}, \mathbf{R}^{-1}(\mathbf{R}\mathbf{U})) = W(\mathbf{x}, \mathbf{U})$$

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This means the strain energy is independent of the rotational part of the deformation.

Implications for the Strain Energy (2/2)

Conclusion

The strain energy depends on $\mathbf{F}$ only through the stretch part $\mathbf{U}$.

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Conclusion

The strain energy depends on $\mathbf{F}$ only through the stretch part $\mathbf{U}$. It's more convenient to use the **Right Cauchy-Green deformation tensor**, $\mathbf{C} = \mathbf{F}^T \mathbf{F} = \mathbf{U}^2$. The final form is:

$$W(\mathbf{x}, \mathbf{F}) = \mathcal{U}(\mathbf{x}, \mathbf{C})$$

Here, $\mathcal{U}$ is an auxiliary function that defines the material's properties.

Stress Tensors & Angular Momentum

Piola-Kirchhoff Stress Tensors

The relationship $W(\mathbf{F}) = \mathcal{U}(\mathbf{C})$ allows us to define two useful stress tensors.

- **1st Piola-Kirchhoff Stress:** $T_{ij} = \frac{\partial W}{\partial F_{ij}}.$
- **2nd Piola-Kirchhoff Stress:** $S_{ij} = 2 \frac{\partial \mathcal{U}}{\partial C_{ij}}.$

They are related by $\mathbf{T} = \mathbf{F}\mathbf{S}$.

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Conservation of Angular Momentum

Because $\mathbf{S}$ is symmetric by definition, the condition $\mathbf{T} = \mathbf{F}\mathbf{S}$ is equivalent to the conservation of angular momentum, which is therefore automatically satisfied.

Linearised Equations of Motion

Our goal is to study small vibrations (like seismic waves) around an equilibrium state.

- We assume the body starts in a **stress-free equilibrium configuration**, where $\varphi^0(\mathbf{x}) = \mathbf{x}$ and the initial stress $\mathbf{T}^0$ is zero.

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- We assume the body starts in a **stress-free equilibrium configuration**, where $\varphi^0(\mathbf{x}) = \mathbf{x}$ and the initial stress $\mathbf{T}^0$ is zero.
- We consider a small displacement $\mathbf{u}(\mathbf{x}, t)$ from this state, seeking a solution of the form:

$$\varphi_i(\mathbf{x}, t) = x_i + su_i(\mathbf{x}, t) + O(s^2)$$

Hooke's Law

We expand the stress tensor for small displacements. Since the initial stress is zero, the first-order term in the expansion is:

$$T_{ij} = s A_{ijkl}(\mathbf{x}) \frac{\partial u_k}{\partial x_l} + O(s^2)$$

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The Elastic Tensor

The quantity **A** is the **elastic tensor**, defined as:

$$A_{ijkl}(\mathbf{x}) = \frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}}(\mathbf{x}, \mathbf{I})$$

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Hyperelastic Symmetry

From the equality of mixed partials, the elastic tensor has a crucial symmetry:

$$A_{ijkl} = A_{klij}$$

The Linearised Wave Equation

Substituting the linearised expressions for velocity and stress into the full equation of motion gives the fundamental equation for elastic waves.

Linearised Equation of Motion

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0$$

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Linearised Boundary Condition

The traction-free boundary condition becomes:

$$n_j A_{ijkl} \frac{\partial u_k}{\partial x_l} = 0$$

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Material Symmetry Group

A material has a symmetry described by a rotation matrix $\mathbf{Q}$ if its strain energy is unchanged by that rotation:

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- **Isotropic:** The material looks the same in all directions.
- **Anisotropic:** The material properties depend on direction.

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Isotropic Elastic Tensor

This symmetry simplifies the elastic tensor to a form with just two constants, the **Lamé parameters** λ and μ :

$$A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{kj})$$

The parameter μ is the **shear modulus**.

Isotropic Materials (2/2)

Isotropic Wave Equation

For a **homogeneous** isotropic material (where ρ, λ, μ are constant), the linearised equation of motion becomes the familiar wave equation:

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Context

This requires assumptions of:

- Small (linearised) motion
- Stress-free initial state
- Isotropy & Homogeneity

Other Material Types

Transversely Isotropic Materials

These materials have a preferred direction (an axis of symmetry, $\hat{\nu}$). They are symmetric with respect to any rotation about that axis.

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Elastic Fluids

Our theory also applies to compressible, non-viscous fluids (e.g., the Earth's outer core).

- A fluid has a preferred volume, but not a preferred shape.
- This is equivalent to an isotropic material with **zero shear modulus**.

Summary (1/2): Key Concepts

- **Frame Indifference:** Understand the polar decomposition theorem ($\mathbf{F} = \mathbf{R}\mathbf{U}$) and how it restricts the strain energy to depend only on the stretch, represented via $\mathbf{C} = \mathbf{F}^T \mathbf{F}$.

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- **Frame Indifference:** Understand the polar decomposition theorem ($\mathbf{F} = \mathbf{R}\mathbf{U}$) and how it restricts the strain energy to depend only on the stretch, represented via $\mathbf{C} = \mathbf{F}^T \mathbf{F}$.
- **Linearisation:** Know the concept of a stress-free equilibrium and the process of linearising the equations of motion for small displacements ($\mathbf{u}$).

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- **Frame Indifference:** Understand the polar decomposition theorem ($\mathbf{F} = \mathbf{R}\mathbf{U}$) and how it restricts the strain energy to depend only on the stretch, represented via $\mathbf{C} = \mathbf{F}^T\mathbf{F}$.
- **Linearisation:** Know the concept of a stress-free equilibrium and the process of linearising the equations of motion for small displacements ($\mathbf{u}$).
- **Material Symmetries:** Understand the physical meaning of the material symmetry group and know the definitions of:
 - Isotropic & Anisotropic
 - Elastic Fluid (zero shear modulus)

Summary (2/2): Key Equations

The Linearised Equation of Motion

You must be able to derive and state this equation:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0$$

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The Elastic Tensor ($\mathbf{A}$)

This tensor contains the material properties. Remember its most important property, the **hyperelastic symmetry**:

$$A_{ijkl} = A_{klij}$$

This symmetry is a direct consequence of the existence of a strain energy function W .